16 Shape **A** and shape **B** are two similar solid shapes.

The ratio of the volume of shape **A** to the volume of shape **B** is 1 : 10

Aisha says that the height of shape **B** is 10 times the height of shape **A**.

(a) Is Aisha correct?
Give a reason for your answer.

No. The height factor is the cube root of 10,

about 2.154, since (height factor)^3 = 10.

1 mark
Use a linear scale factor.

(1)

The surface area of shape **B** is 72 cm^2

(b) Work out the surface area of shape **A**.
Give your answer correct to 3 significant figures.

Linear factor B to A = $10^{-1/3}$

Area factor B to A = $10^{-2/3}$

1 mark
Square the linear factor.

Area A = 72 ÷ (cube root of 10)^2

= 15.511929... cm^2 = 15.5 cm^2 (3 s.f.)

1 mark
15.5 cm^2.

..... cm^2

(2)

(Total for Question 16 is 3 marks)